Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

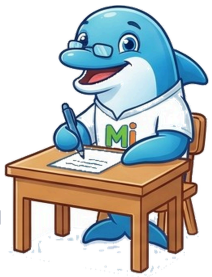
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Edition 1, 2026. First printing.

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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big Number Puzzles	6-7	8-9	10 and 12	11	☐
3	Operation detectives	Not in this proof edition				☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal detectives	Not in this proof edition				☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and nets	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

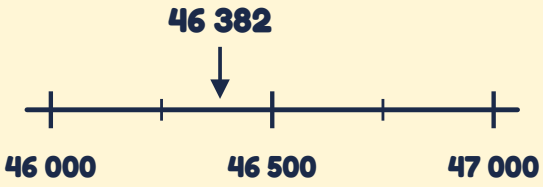
Learn it: Numbers to 100 000

My goal. I can round numbers to the nearest 10, 100 and 1000. I can work backwards from a rounded number and solve digit puzzles.

To round a number, find the two neighbours it lies between. Halfway rounds up. To work backwards from a rounded number, find the smallest and the largest numbers that round to it.

Worked example 1

Round 46 382 to the nearest 1000.

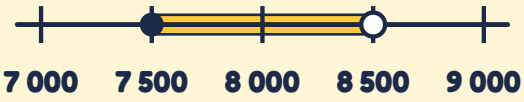


1. 46 382 is between 46 000 and 47 000.
2. The halfway mark is 46 500. 46 382 is before it.

Answer: 46 382 rounds to 46 000.

Worked example 2

A number rounds to 8000 to the nearest 1000. What are the smallest and the largest it can be?



1. 7 500 is halfway, so it rounds up to 8000.
2. 8 500 rounds up to 9000, so the last number is 8 499.

Answer: Smallest 7 500 and largest 8 499.

How to round

1. Find the 2 neighbours, such as 46 000 and 47 000.
2. Find the halfway number. Halfway rounds up.
3. To work backwards, find where the rounding changes.

Place value

10 000s	1000s	100s	10s	1s
3	5	2	0	6

- 35 206 = 30 000 + 5000 + 200 + 6

Your turn

Answer at the back of the book.

Try **Round 52 718 to the nearest 100.** ☐

Start here

Do these first. Take your time.

1 Round each number. Write the answers in the table.

Number	Nearest 10	Nearest 100	Nearest 1000
38 746			
70 049			
9996			

2 Write the value of the digit 6 in each number.

(a) 46 281: ☐ (b) 60 912: ☐ (c) 83 062: ☐

3 Write each number in figures.

In words	In figures
forty-two thousand and seven	
sixty thousand, nine hundred	
eight thousand and thirty	

4 Write the numbers in order. Begin with the greatest.

58 304 58 340 58 034 58 430

_____

Start here

Keep going. Check your work.

5 The arrows point to numbers on the line. Each small step is 100.



(a) A is ☐

(b) B is ☐

(c) Which thousand is B closer to? ☐

6 A match had 36 482 people watching. Round the number of people to the nearest 1000 and to the nearest 100.

(a) Nearest 1000: ☐ (b) Nearest 100: ☐

7 A town has 12 000 people after rounding to the nearest 1000. Find the least number of people it could have, and the greatest.

(a) Least: ☐ (b) Greatest: ☐

Working: _____

8 A number rounds to 4700 to the nearest 100.

(a) The smallest it can be is ☐ (b) The largest it can be is ☐

(c) How many whole numbers could it be? ☐

Start here

Nearly there. You can do it.

9 **A ferry carries 4018 people on each trip. It makes 5 trips.**

(a) Round 4018 to the nearest 1000. Estimate the total: ☐

(b) Find the exact total: ☐

(c) The exact total is more less than the estimate.

Working: _____

10 **I have 5 digits. Round me to the nearest 1000 and you get 52 000. My hundreds digit is 7. My tens digit is 5 and my ones digit is 2. Who am I?**

I am ☐.

11 **Use the digit cards 2, 4, 0, 7 and 9. Use each card once.**

(a) The greatest number: ☐

(b) The least number: ☐

(c) The greatest odd number: ☐

12 **Find the rule. Write the next 2 numbers.**

(a) 31 600, 32 600, 33 600, ☐, ☐

(b) 70 400, 68 900, 67 400, ☐, ☐

(c) 4997, 5000, 5003, 5006, ☐, ☐

13 **Circle the correct sign.**

(a) 45 009 > < 45 090

(b) 8999 > < 10 000

(c) 62 100 > < 62 010

(d) 30 000 + 700 > < 30 070

Think harder

Hints are upside down. Try first!

1 **Cards: 1, 3, 4, 6, 8. Use each card once. Make the 5-digit number nearest to 40 000.**

☐

Working: _____

_____

_____

Hint. Check one number just below 40 000 and one just above.

2 **Two crowds were counted. To the nearest 1000, one crowd was 41 000 and the other was 28 000.**

(a) The smallest possible total of both crowds: ☐

(b) The largest possible total of both crowds: ☐

Working: _____

_____

Hint. Add the 2 smallest numbers. Then add the 2 largest.

3 **A 5-digit number has 5 different digits. It rounds to 70 000 to the nearest 10 000.**

(a) The greatest it can be: ☐

(b) The least it can be: ☐

Working: _____

_____

Hint. Numbers that round to 70 000 run from 65 000 to 74 999.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Ravi rounds 36 449 in 3 steps. First to the nearest 10, then that answer to the nearest 100, then that answer to the nearest 1000.**

(a) After the 3 steps he gets ☐

(b) Round 36 449 to the nearest 1000 in one go: ☐

(c) Rounding 1 step at a time gives a wrong answer for some 4-digit numbers.
What is the least 4-digit number where it does? ☐

Hint. Which numbers end in 4 and 5 just before a change?

2 **Beyond Level 4: Counting by place**
Ben writes the whole numbers from 1 to 1000. How many of them contain at least one 7?

☐

Working: _____

Hint. Count the numbers with no 7. Choose each digit in turn.

3 **Two patterns are woven together: 12 000, 3, 11 000, 6, 10 000, 9, and so on.**

(a) The 9th number is ☐

(b) The 10th number is ☐

Hint. Take every second number. You get 2 patterns.

4 **Some numbers round to 7000 to the nearest 1000. Siti says they all round to 7000 to the nearest 100 too.**

(a) Is she right? Yes No

(b) How many of the whole numbers that round to 7000 to the nearest 1000 also round to 7000 to the nearest 100? ☐

Hint. Find the first and last numbers for each rounding.

✔ Check yourself

No hints. Try each one on your own.

1 **Round 25 962. To the nearest 10: ☐ 100: ☐ 1000: ☐**

Working: _____

2 **To the nearest 100, 8300 people came to a show. The least could be ☐ and the greatest could be ☐.**

Working: _____

3 **Cards: 4, 0, 7, 1, 8. Use each card once. Make the least odd 5-digit number.** ☐

Working: _____

4 **Find the rule. Write the next 2 numbers. 53 200, 51 700, 50 200, ☐, ☐**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big Number Puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 2 · Big Number Puzzles

Your turn
page 6

Answer: 52 700
Why: 52 718 is between 52 700 and 52 800. The halfway mark is 52 750. 52 718 is before it, so it rounds down to 52 700.

START HERE

1
page 7

Answer: 38 750, 38 700, 39 000, 70 050, 70 000, 70 000, 10 000, 10 000, 10 000
Why: 38 746: the 6 ones round up to 38 750. To the nearest 100, 46 is under 50, so 38 700. To the nearest 1000, 746 is over 500, so 39 000. 70 049: 70 050, 70 000, 70 000. 9996 is 4 away from 10 000, so all three roundings are 10 000.

2
page 7

Answer: (a) 6000 (b) 60 000 (c) 60
Why: 46 281 has 6 in the thousands place, so it is worth 6000. In 60 912 it is in the ten thousands place: 60 000. In 83 062 it is in the tens place: 60.

3
page 7

Answer: 42 007, 60 900, 8030
Why: forty-two thousand and seven is 42 007: 42 thousands, no hundreds, no tens and 7 ones. sixty thousand, nine hundred is 60 900. eight thousand and thirty is 8030.

4
page 7

Answer: 58 430, 58 340, 58 304, 58 034
Why: All four have 58 thousands. Compare the hundreds: 4, 3, 3, 0. The two with 3 hundreds are 58 340 and 58 304, and 340 is greater than 304.

5
page 8

Answer: (a) 24 300 (b) 24 800 (c) 25 000
Why: There are 5 steps of 100 between 24 000 and 24 500. A is 3 steps after 24 000: 24 300. B is 3 steps after 24 500: 24 800. 24 800 is past the halfway mark 24 500, so it is closer to 25 000.

6
page 8

Answer: (a) 36 000 (b) 36 500
Why: 36 482 is between 36 000 and 37 000, and it is before the halfway mark 36 500, so it rounds to 36 000. To the nearest 100 it is between 36 400 and 36 500, and 82 is more than 50, so 36 500.

7
page 8

Answer: (a) 11 500 (b) 12 499
Why: 11 500 is halfway between 11 000 and 12 000 and rounds up to 12 000. 12 500 would round up to 13 000, so the largest is 12 499.

8
page 8

Answer: (a) 4650 (b) 4749 (c) 100
Why: 4650 is halfway between 4600 and 4700, so it rounds up. 4750 would round to 4800, so the largest is 4749. From 4650 to 4749 there are 100 numbers.

9
page 9

Answer: (a) 20 000 (b) 20 090 (c) more
Why: 4018 rounds to 4000, and 5 × 4000 = 20 000. The exact total is 5 × 4018 = 20 090. 4018 was rounded down, so the exact total is more.

10 page 9

Answer: 51 752
Why: Numbers that round to 52 000 are from 51 500 to 52 499. With 7 hundreds the number must be 51 7__, because 52 7__ is too big. Then the tens and ones give 51 752.

11 page 9

Answer: (a) 97 420 (b) 20 479 (c) 94 207
Why: Greatest: put the greatest digits first, 97 420. Least: 0 cannot go first, so start with 2: 20 479. Odd: the ones digit must be 7 or 9. Ending in 7 gives 94 207 and ending in 9 gives 74 209, so 94 207 is greater.

12 page 9

Answer: (a) 34 600, 35 600 (b) 65 900, 64 400 (c) 5009, 5012
Why: (a) Add 1000 each time. (b) Take away 1500 each time: 67 400 – 1500 = 65 900, then 64 400. (c) Add 3 each time: 5006 + 3 = 5009, then 5012.

13 page 9

Answer: (a) < (b) < (c) > (d) >
Why: (a) 45 009 has 0 tens, 45 090 has 9 tens, so it is smaller. (b) 8999 has 4 digits. (c) 62 100 has 1 hundred, 62 010 has none. (d) 30 000 + 700 is 30 700, which is greater than 30 070.

THINK HARDER

1 page 10

Answer: 38 641
Why: Just below 40 000: the greatest number that starts with 3 is 38 641, which is 1359 away. Just above 40 000: the least number that starts with 4 is 41 368, which is 1368 away. 38 641 is closer.

2 page 10

Answer: (a) 68 000 (b) 69 998
Why: The smallest crowds are 40 500 and 27 500: 68 000. The largest are 41 499 and 28 499: 69 998.

3 page 10

Answer: (a) 74 986 (b) 65 012
Why: Greatest: 74 999 is the top, but its digits repeat. Using 7, 4, then the greatest digits left gives 74 986. Least: 65 000 repeats the 0, so use 65 012.

GO FURTHER

1 page 11

Answer: (a) 37 000 (b) 36 000 (c) 1445
Why: 36 449 becomes 36 450, then 36 500, then 37 000. In one go it is 36 000, because 449 is below 500. For part (c), 1445 becomes 1450, then 1500, then 2000, but in one go it is 1000. Any 4-digit number below 1445 gives the same answer both ways.

2 page 11

Answer: 271
Why: Count the numbers with no 7. Think of 3 digits, from 000 to 999. Each digit can be any of 9 digits (not 7): 9 × 9 × 9 = 729. That includes 000, which is not in the list, and it leaves out 1000, which has no 7, so 729 numbers from 1 to 1000 have no 7. So 1000 – 729 = 271 have a 7.

3 page 11

Answer: (a) 8000 (b) 15
Why: The numbers in places 1, 3, 5 ... are 12 000, 11 000, 10 000, 9000, 8000. The 9th is the 5th of these: 8000. The numbers in places 2, 4, 6 ... are 3, 6, 9, 12, 15. The 10th is the 5th of these: 15.

4
page 11

Answer: (a) No (b) 100
Why: To the nearest 1000, the numbers are 6500 to 7499. To the nearest 100, they are 6950 to 7049. For example, 7400 is in the first list but rounds to 7400 to the nearest 100. The numbers in both lists are 6950 to 7049: 100 numbers.

CHECK YOURSELF

1
page 12

Answer: 25 960, 26 000, 26 000
Why: 25 962: the 2 ones round down to 25 960. The last two digits, 62, are over 50, so 26 000. The last three digits, 962, are over 500, so 26 000.

2
page 12

Answer: Least 8250, greatest 8349
Why: 8250 is halfway between 8200 and 8300 and rounds up to 8300. 8350 would round up to 8400, so the greatest is 8349.

3
page 12

Answer: 10 487
Why: The ones digit must be 7 or 1. Ending in 7 lets 1 go first: 10 487. Ending in 1 needs 4 first: 40 781. 10 487 is less.

4
page 12

Answer: 48 700, 47 200
Why: Each number is 1500 less than the one before: 50 200 – 1500 = 48 700, then 47 200.